

Roll Number:

Name:

Section:



Indian Institute of Information Technology, Sri City, Chittoor
Monsoon Semester Schedule Quiz 1, August-2026
UG-2, Third Sem

Duration: 20 Minutes

Name of the Exam: RANAC (S1)

Max. Marks: 10

August 28, 2026

Instructions:

1. Calculators are allowed.
2. All questions are mandatory and carry equal mark.
3. Tick only the right option. If we find two ticks for one question, '0' marks will be awarded.
4. Rough work should not be done in the question paper.
5. Return the question paper along with the rough sheet before leaving the exam hall.

1. The following data are given for a polynomial $P(x)$ of unknown degree:

x	0	1	2
$P(x)$	2	-1	4

Determine the coefficient of x^3 in $P(x)$ if all third-order forward differences are 1.

- (a) 6
 - (b) 3
 - (c) $1/3$
 - (d) $1/6$
2. For a polynomial $P_n(x)$ interpolating $n+1$ distinct points, which statement is correct?
- (a) The polynomial is always exactly of degree n .
 - (b) The polynomial has degree at most n .
 - (c) The polynomial has degree at least n .
 - (d) Its degree cannot be determined.
3. Suppose the data are $(0,0)$, $(1,1)$, $(3,9)$. The interpolation points are not equally spaced. Which method is most directly applicable?
- (a) Newton's forward finite-difference interpolation
 - (b) Newton's backward finite-difference interpolation
 - (c) Newton's Divided-difference interpolation
 - (d) Centered interpolation
4. The central difference operator δ is related to the shift operator by
- (a) $\delta = E - 1$

- (b) $\delta^2 = E + E^{-1} - 2$
 (c) $\delta = E - E^{-1}$
 (d) $\delta = E^{1/2} + E^{-1/2}$

5. Let $\{a_n\}$ be a sequence of real numbers. Which of the following statements is true?

- (a) If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty} a_n^4$ is convergent.
 (b) If $\sum_{n=1}^{\infty} |a_n|$ is convergent, then $\sum_{n=1}^{\infty} a_n^2$ is convergent.
 (c) If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\sum_{n=1}^{\infty} a_n^4$ is divergent.
 (d) If $\sum_{n=1}^{\infty} |a_n|$ is divergent, then $\sum_{n=1}^{\infty} a_n^2$ is divergent.

6. For $x > 0$, determine the condition on x for the convergence of

$$\sum_{m=2}^{\infty} x^{\log m},$$

where $\log$ denotes the natural logarithm.

- (a) $0 < x < e^{-1}$
 (b) $0 < x < 1$
 (c) $x > e$
 (d) $0 < x < e$

7. Discuss the convergence of the series

$$\sum_{n=3}^{\infty} \frac{1}{(n \log n)(\log \log n)^p}.$$

- (a) Converges for $p > 0$
 (b) Converges for $p > 1$
 (c) Converges for $p \geq 1$
 (d) Diverges for all p

8. $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \right)$ is equal to

- (a) $\log 2$
 (b) 0
 (c) 1
 (d) $\log \left(\frac{1}{2} \right)$

9. $\lim_{n \rightarrow \infty} \int_1^2 e^{-nx^2} dx$

- (a) 1
 (b) 2
 (c) 0
 (d) Does not exist finitely

10. The sequence

$$a_n = \frac{1}{2} \left(a_{n-1} + \frac{2025}{a_{n-1}} \right), \quad a_1 = 2026,$$

is convergent. Then

is

$$\lim_{n \rightarrow \infty} a_n$$

- (a) 43
- (b) 45
- (c) 46
- (d) Does not exist
